Ben Wilfong

1 Basic Information

Title: Graduate Research Assistant
Citizenship: United States of America

Georgia Institute of Technology
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Research Interests: Computational fluid dynamics, multiphase fluid dynamics, high-performance computing, GPU-accelerated computing, distributed computing, numerical methods for PDEs, scientific computing

2 Education

Georgia Institute of Technology
(In Progress) Doctor of Philosophy, Computational Science and Engineering
Advisor: Dr. Spencer Bryngelson

Georgia Institute of Technology
(2026) Master of Science, Computational Science and Engineering

Rose–Hulman Institute of Technology
(2022) Bachelor of Science, Mechanical Engineering and Computational Science

3 Experience

PSAAP IV Student Scholar — May 2026 – August 2026

Institution: Lawrence Livermore National Laboratory
Supervisor: Dr. Britton Olson
Duties: Implementation, validation, and analysis of regularization methods for high-order compact finite difference schemes for compressible, multiphase fluid dynamics simulations.

Defense Science Technology Intern — June 2022 – August 2022

Institution: Lawrence Livermore National Laboratory
Supervisor: Dr. Kyle Sinding
Duties: Perform finite element simulations using LLNL's HPC resources to run ALE3D, generate case files and input data, post-process data to gather useful quantities of interest like shock wave arrival time and pressure profiles

EERE High Performance Computing for Manufacturing Intern — June 2021 – August 2021

Institution: Lawrence Livermore National Laboratory in collaboration with Oak Ridge Institute for Science and Education (ORISE)
Supervisor: Dr. John Karnes
Duties: Perform molecular dynamics simulations using LLNL's HPC resources using LAMMPS, generate case files and input data, post-process data to gather useful quantities of interest like reaction rates and percent conversion

4 Awards

- (2025) ACM Gordon Bell Prize Finalist (*Simulating many-engine spacecraft: Exceeding 1 quadrillion degrees of freedom via information geometric regularization*)
- (2025) OLCF User Meeting Best Data Visualization Award 1st Place
- (2024) CRNCH Fellowship for Novel Computing Paradigms and Hierarchies
- (2022) Georgia Tech President's Fellowship
- (2018-22) Dean's List, Rose–Hulman Institute of Technology

5 Professional activity

5.1 Memberships

- (2024-Present) Association for Computing Machinery (ACM), Member
- (2024-Present) American Physical Society (APS), Member

6 Service and outreach

- (2025) SC25 student volunteer
- (2023-Present) President's undergraduate research award reviewer
- (2023-24) CSE Graduate Student Association vice-president
- (2022-23) CSE Graduate Student Association events chair

7 Media

The following media mentions work that I have made significant contributions to.

- (2025) Record-Breaking Simulation Boosts Rocket Science and Supercomputing to New Limits [LINK]
- (2025) Gordon Bell finalist team pushes scale of rocket simulation on El Capitan [LINK]
- (2025) It is Rocket Science [LINK]
- (2025) Fluid Flow Simulation on Frontier earns Gordon Bell Finalist Selection [LINK]
- (2025) Shock Treatment for CFD Simulations [LINK]
- (2025) The OLCF's Problem Busters [LINK]
- (2023) Group Optimizes Fluid Dynamics Simulator on World's Fastest Supercomputer [LINK]

8 Publications

8.1 Preprints

[PP1] T. Chu, B. Wilfong, T. Koehler, R. M. McMullen, and S. H. Bryngelson (2026a). "Coupled Rayleigh–Taylor and Faraday instabilities in vertically vibrated cylindrical containers". arXiv:2607.28932. DOI: 10.48550/arXiv.2607.28932

8.2 Archival, heavily refereed papers

[P5] A. Radhakrishnan, B. Wilfong, S. H. Bryngelson, and F. Schäfer (2026). "Shocks without shock capturing: Information geometric regularization of finite volume methods for Navier-Stokes-like problems". In: *Computer Methods in Applied Mechanics and Engineering* 461, p. 119216. DOI: 10.1016/j.cma.2026.119216

[P4] D. Vaca-Revelo, B. Wilfong, S. H. Bryngelson, and A. Gnanaskandan (2026). "Hardware-accelerated phase-averaging for cavitating bubbly flows". In: *International Journal of Multiphase Flow*, p. 105674. ISSN: 0301-9322. DOI: 10.1016/j.ijmultiphaseflow.2026.105674

[P3] B. Wilfong, H. Le Berre, A. Radhakrishnan, A. Gupta, D. J. Vickers, D. Vaca-Revelo, D. Adam, H. Yu, H. Lee, J. R. Chreim, M. Carcana Barbosa, Y. Zhang, E. Cisneros-Garibay, A. Gnanaskandan, M. Rodriguez Jr., R. D. Budiardja, S. Abbott, T. Colonius, and S. H. Bryngelson (2026). "MFC 5.0: An exascale many-physics flow solver". In: *Computer Physics Communications* 322, p. 110055. DOI: 10.1016/j.cpc.2026.110055

[P2] T. Chu, B. Wilfong, T. Koehler, R. M. McMullen, and S. H. Bryngelson (2025). "Competing mechanisms at vibrated interfaces of density-contrast fluids". In: *Physical Review Fluids* 10 (9), p. 093904. DOI: 10.1103/r9b3-psg4

[P1] A. Radhakrishnan, H. Le Berre, B. Wilfong, J.-S. Spratt, M. Rodriguez Jr., T. Colonius, and S. H. Bryngelson (2024). "Method for portable, scalable, and performant GPU-accelerated simulation of multiphase compressible flow". In: *Computer Physics Communications* 302, p. 109238. DOI: 10.1016/j.cpc.2024.109238

8.3 Conference papers

[C5] T. Chu, B. Wilfong, T. Koehler, R. M. McMullen, and S. H. Bryngelson (2026b). "Linear and numerical analysis of interfacial instabilities in vibrated cylindrical containers". In: *AIAA AVIATION 2026 Forum*, p. 4363. DOI: 10.2514/6.2026-4363

[C4] B. Wilfong, A. Radhakrishnan, H. Le Berre, D. J. Vickers, T. Prathi, N. Tselepidis, B. Dorschner, R. Budiardja, B. Cornille, S. Abbott, F. *Schäfer, and S. H. *Bryngelson (2025). "Simulating many-engine spacecraft: Exceeding 1 quadrillion degrees of freedom via information geometric regularization". In: *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*. *Equal contribution. DOI: 10.1145/3712285.3771783

[C3] B. Wilfong, A. Radhakrishnan, H. A. Le Berre, T. Prathi, S. Abbott, and S. H. Bryngelson (2025). "Testing and benchmarking emerging supercomputers via the MFC flow solver". In: *Proceedings of the SC '25 Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis*. DOI: 10.1145/3731599.3767424

[C2] B. Wilfong, A. Radhakrishnan, H. A. Le Berre, S. Abbott, R. D. Budiardja, and S. H. Bryngelson (2024). "OpenACC offloading of the MFC compressible multiphase flow solver on AMD and NVIDIA GPUs". In: *Proceedings of the SC '24 Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis*. DOI: 10.1109/SCW63240.2024.00242

[C1] B. A. Wilfong, R. McMullen, T. Koehler, and S. H. Bryngelson (2024). "Instability of Two-Species Interfaces via Vibration". In: *AIAA AVIATION FORUM AND ASCEND 2024*. DOI: 10.2514/6.2024-4480

8.4 Conference abstracts

Presenter indicated by asterisk (*).

[A15] B. Wilfong*, T. Chu, T. Koehler, R. McMullen, and S. H. Bryngelson (2026). "The impact of acceleration magnitude and frequency on gas transport in vibrated bubbly flow". In: *79th Annual Meeting of the APS Division of Fluid Dynamics (APS DFD)*

[A14] T. Chu*, B. Wilfong, T. Koehler, R. McMullen, and S. H. Bryngelson (2026). "Coupled Rayleigh-Taylor and Faraday instabilities in vibrated cylindrical containers". In: *79th Annual Meeting of the APS Division of Fluid Dynamics (APS DFD)*

[A13] J. Gillis*, B. Wilfong, D. Vickers, S. H. Bryngelson, T. Jackson, and S. Balachandar (2026). "Toward predictive simulation of particle instabilities in explosive dispersal using the exascale multiphysics flow solver MFC". in: *79th Annual Meeting of the APS Division of Fluid Dynamics (APS DFD)*

[A12] B. C. Kim*, M. Vandenboom, T. Jackson, S. H. Bryngelson, B. Wilfong, D. Vickers, and S. Balachandar (2026). "Database construction and characterization of pseudo-turbulence closure quantities for stationary particle arrays in compressible flow". In: *79th Annual Meeting of the APS Division of Fluid Dynamics (APS DFD)*

[A11] S. H. Bryngelson*, A. Radhakrishnan, and B. Wilfong (2026). "Extreme-scale simulation of many-engine spacecraft via regularization". In: *US National Congress on Theoretical and Applied Mechanics*

[A10] S. H. Bryngelson*, A. Radhakrishnan, B. Wilfong, and F. Schäfer (2026). "Shocks without tracking or capturing: Exceeding 1 quadrillion degrees of freedom via inviscid regularization of the compressible Navier-Stokes equations". In: *APS Global Physics Summit*

[A9] S. H. Bryngelson*, T. Prathi, A. Radhakrishnan, D. Vickers, and B. Wilfong (2026). "Compressible flow with a free lunch: Simulating 1 quadrillion degrees of freedom via regularization without loss of accuracy". In: *SIAM Conference on Parallel Processing for Scientific Computing (PP26)*

[A8] B. Wilfong*, T. Chu, R. McMullen, T. Koehler, and S. H. Bryngelson (2025). "Euler–Lagrange simulations of near-surface gas transport in vibrated bubbly flows". In: *78th Annual Meeting of the APS Division of Fluid Dynamics (APS DFD)*

[A7] T. Chu*, B. Wilfong, T. Koehler, R. McMullen, and S. H. Bryngelson (2025). "Competing mechanisms at vibrated interfaces of density-contrast fluids". In: *78th Annual Meeting of the APS Division of Fluid Dynamics (APS DFD)*

[A6] S. H. Bryngelson*, B. Wilfong, A. Radhakrishnan, and F. Schäfer (2025). "Simulating shock-laden and multiphase flows interface without tracking or dissipation". In: *78th Annual Meeting of the APS Division of Fluid Dynamics (APS DFD)*

[A5] S. H. Bryngelson*, B. Wilfong, and A. Radhakrishnan (2025). "Numerics for diverse exascale platforms with application to multi-phase and species flow". In: *18th U.S. National Congress on Computational Mechanics (USNCCM)*

[A4] A. Radhakrishnan, H. Le Berre, B. Wilfong, R. Budiardja, S. Abbott, and S. H. Bryngelson* (2025). "Compressible flow simulations on Frontier and El Capitan architectures". In: *APS Global Physics Summit*

[A3] S. H. Bryngelson*, A. Radhakrishnan, and B. Wilfong (2025). "Solving Navier–Stokes-like problems with discontinuous features via information geometric regularization". In: *3rd Virtual UNCG PDE Conference of 2025*

[A2] B. Wilfong*, T. Chu, R. McMullen, T. Koehler, and S. H. Bryngelson (2024). "Hydrodynamic instability and breakup of a liquid-gas interface via vibration". In: *77th Annual Meeting of the APS Division of Fluid Dynamics (APS DFD)*

[A1] B. Wilfong, A. Radhakrishnan, and S. H. Bryngelson* (2024). "Multiphase flow numerics: Perspectives from exascale simulation". In: *5th International Conference on Numerical Methods for Multiphase Flow (ICNMMF)*

8.5 Other contributions

[O2] B. Wilfong, T. Chu, R. McMullen, T. Koehler, and S. H. Bryngelson (2026). "From ripples to rain: Faraday waves and droplet ejection in a vertically vibrated liquid bath". In: *79th Annual Meeting of the APS Division of Fluid Dynamics (APS DFD)*. (poster)

[O1] D. Vickers, B. Wilfong, and S. H. Bryngelson (2026). "The irregularity of particle-laden flow: Forming gas jets in a hemispherical shell". In: *79th Annual Meeting of the APS Division of Fluid Dynamics (APS DFD)*. (video)

9 Skills

- *Programming Languages/Paradigms:* Fortran, OpenACC, OpenMP, MPI, MATLAB, Bash
- *Software and Tools:* ParaView, VisIt, SolidWorks, SLURM, PBS, Nsight Systems, Nsight Compute, RocProf, Omniperf, Fypp